Profiles & bond energies

Follow energy transfer

1 Energy is conserved

Exothermic pays energy out to surroundings → surroundings warm; products have less energy.

Endothermic takes energy from surroundings → surroundings cool; products have more energy.

A starting spark does not make an overall reaction endothermic.

2 Examples & applications

Exothermic: combustion, many oxidation reactions, neutralisation → hand warmers/self-heating cans.

Endothermic: thermal decomposition; citric acid + sodium hydrogencarbonate → cooling applications.

Judge temperature, duration, safety, cost and disposal.

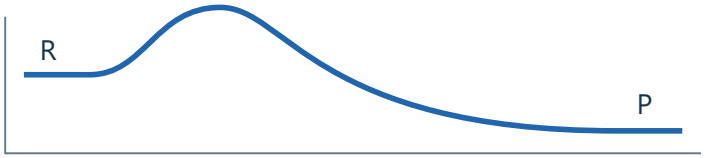
Read the profile

3 Profile axes & peak

Energy vertically; reaction progress horizontally. Reactants and products at flat levels joined by curved peak.

Activation energy: minimum collision energy; measure reactant level to peak. Overall change is between reactant and product levels.

4 Products decide the sign



Exothermic: products (P) lower

Products lower → energy transferred out, negative overall change. Products higher → energy taken in, positive.

Activation energy is not measured from the baseline; it is not the overall change.

Both exothermic and endothermic reactions need activation energy.

Calculate the balance

5 Count bonds correctly

Overall = broken − formed. Breaking takes in energy; making releases it. Include equation coefficients.

$H_2 + Cl_2 \rightarrow 2HCl$. Given 436, 243, 431 kJ/mol: break 436 + 243 = 679; form 2 × 431 = 862.

679 − 862 = **−183 kJ** for molar quantities in equation: exothermic.

6 Unknown bond energy

If H–Cl energy is x and overall change is −183: **679 − 2x = −183.**

2x = 862 → x = **431 kJ/mol.** Use the data supplied.

CH_4 has four C–H bonds; $2H_2O$ contains four O–H bonds. Do not confuse bonds with molecules.